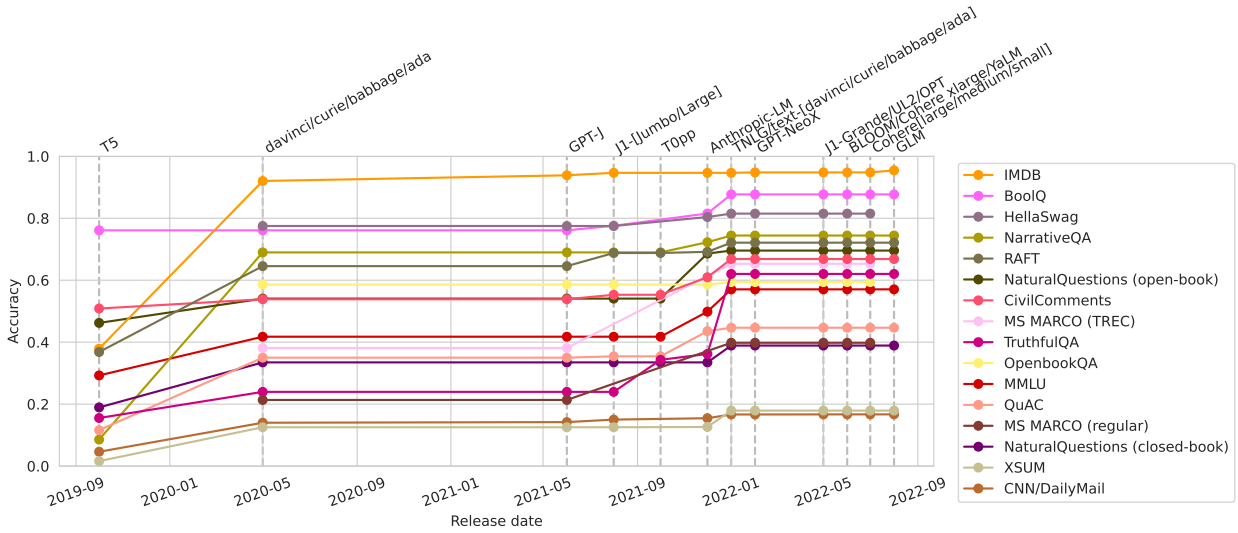

Figure 27: **Cumulative accuracy over time.** The relationship between time (x-axis) and the accuracy of the most accurate model released up to that point (y-axis) across 16 core scenarios. That is, the graph tracks the progress in the state-of-the-art (SOTA) accuracy over time for each scenario.

in the top half of this tier. Similarly, within a model family, we see model scale is perfectly monotonically correlated with model accuracy win rate (e.g. see the OpenAI and Cohere model variants). We emphasize this certainly does not imply the models that fare poorly, or the small models, are inaccurate under all conditions: we expect that the prompting approaches we used are poorly suited for these smaller models and, perhaps, it is better to fine-tune these models (cf. Liu et al., 2022a).

Consistent with previous findings of accuracy being correlated with robustness and fairness, we see similar rankings of models for those scenarios. However, we note that BLOOM (176B) notably performs considerably better (in a relative sense) for robustness/fairness than its accuracy would suggest, with OPT (175B) and GLM (130B) roughly interchanging positions for robustness and accuracy (i.e. GLM (130B) is more robust and less accurate in a relative sense compared to OPT (175B)). In contrast, perhaps also as expected given the weak correlations of generative harms with other metrics, we see the rankings for toxicity and bias are notably quite different. We highlight how T0++ (11B) and davinci (175B) in particular demonstrate the opposite trends when comparing their bias win rates to their toxicity win rates. That is, T0++ (11B) is the most toxic when compared to all other models, but one of the three least (gender) biases, whereas davinci (175B) is one of the four most biased head-to-head, but one of the less toxic models.

Accuracy as a function of other variables. To build on these model comparisons, we now consider trends in model accuracies as a function of other relevant variables. In Figure 27, we report model accuracies as a function of time (i.e. model release date). As a function of time, we see that the release of GPT-3 (Brown et al., 2020) clearly establishes a strong baseline for future models across all scenarios, with a distinctive improvement over T5 (11B). To the extent there is upward movement in accuracy since then, we often see it coming with the introduction of Anthropic-LM v4-s3 (52B) roughly 18 months later in December, 2021 (the first model to use reinforcement learning with human feedback of those we evaluate), though we see a surprisingly clear monotonic improvement over the intermediary period for **NaturalQuestions**.

In Figure 28, we stratify the results for accuracy based on model accessibility (see Liang et al., 2022) as we discuss in §6: MODELS. For each access category, we report the per-scenario accuracy of the model in that category that is the most accurate (full bar) and that is generally the most accurate (smaller dark sub-bar; based on Figure 26). This means the sub-bar is always the accuracy of OPT (175B) for **open** models, text-davinci-002 for **limited access** models, and TNLG v2 (530B) for **closed** models. We see that these models for **limited** access are consistently the best in their categories, but OPT (175B) is sometimes

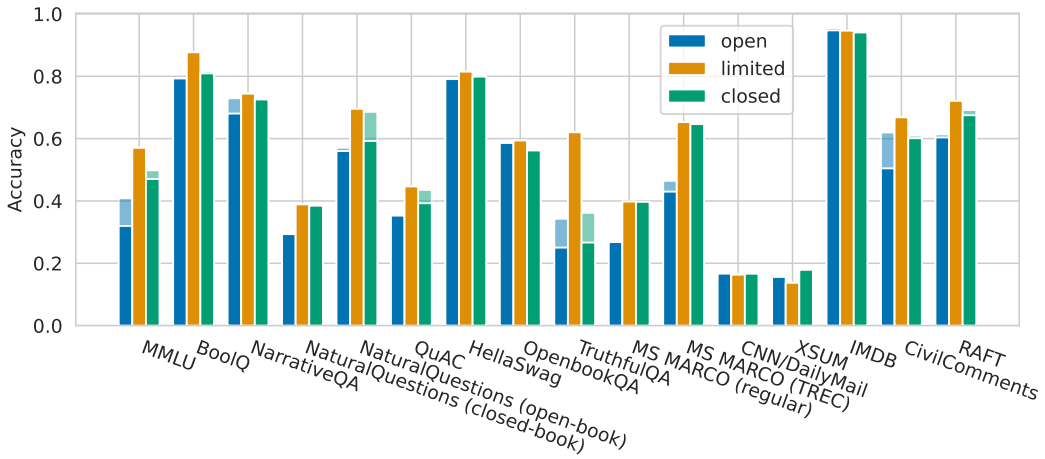

Figure 28: **Accuracy as a function of model access.** The relationship between access (**open** vs. **limited** vs. **closed**) and model accuracy for each of the 16 core scenarios. Shaded bars indicate the performance of the best model for that scenario, whereas the solid bars indicate the performance of the overall most accurate model across all core scenarios based on Figure 26.

improved over (e.g. 10 point improvement for **MMLU** and **CivilComments**) as is TNLG v2 (530B) (e.g. more than 5 point improvement for **NaturalQuestions** (open-book) and **TruthfulQA**).

Overall, we see the accuracies generally fall in the direction **limited access** > **closed** > **open**, which aligns with Figure 26. With that said, we see some reason for optimism regarding open science through open-sourced models, in that the best public model is usually somewhat competitive in accuracy (i.e. within 5 points of the best non-open model), with the notable exceptions of more knowledge-intensive scenarios of **MMLU**, **NaturalQuestions** (closed-book) and **TruthfulQA** as well as both information retrieval scenarios. This is especially noteworthy given we have yet to see models being open-sourced with significant use of human preferences and reinforcement learning from human feedback, which may further bridge this gap. However, we do highlight there are other confounds: some models may not be disclosed to the public (let alone released), some models may be disclosed after significant delay, and some highly accurate models are not yet evaluated in HELM (e.g. Gopher, Chinchilla, PaLM; see §10.4: MISSING-MODELS). We hope our findings on the relationship between model accessiblity and model performance (e.g. accuracy) will inform the development of community norms and standards around model access/release (Liang et al., 2022).

Predicting accuracy. Given the resource-intensive nature of producing language models, especially the most capable models (for which we take accuracy for different scenarios as a proxy), the ability to predict model capabilities[65] is valuable for orienting decision-making. Prior work demonstrates model scale (whether measured in model size, data size, or compute/training FLOPs) reliably predicts model loss for language models, establishing *scaling laws* (Kaplan et al., 2020; Hoffmann et al., 2022). On the other hand, work has also shown that certain qualitative capabilities (e.g. in-context learning; Brown et al., 2020) *emerges* unpredictably beyond a critical scale threshold (see Wei et al., 2022b). And many works have implicitly assumed and demonstrated that improvements in language modeling loss (i.e. perplexity) translate to improvements on downstream use cases. In Figure 29, we report on the relationship between model scale and accuracy, while in Figure 30, we report perplexity (measured on **The Pile**) vs. accuracy.

In light of the very consistent picture that model scale within a family led to reliable improvements in accuracy (Figure 26), we see that this does not generalize across model families. If we are to plot model scale vs. accuracy, we find the relationship is very chaotic; model scale (agnostic to model family) as measured in parameters is a poor predictor of trends in accuracy. Unfortunately, the data on different models was too

[65]It would be similarly interesting to predict our properties (e.g. those related to harms), though here we focus on accuracy. We do note that our evaluation deliberately surfaces all the necessary metrics required to project other properties.

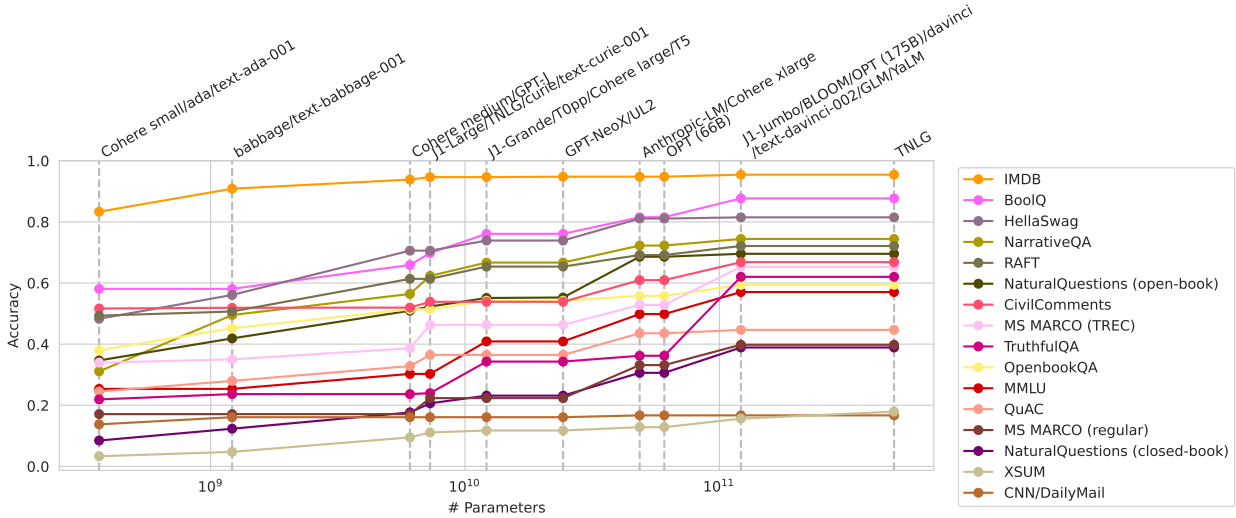

Figure 29: **Cumultative Scale vs. Accuracy.** The relationship between model parameter size (x-axis) and the accuracy of the most accurate model released up to that scale on each core scenario. That is, the graph tracks the progress in the state-of-the-art (SOTA) accuracy as a function of scale for each scenario.

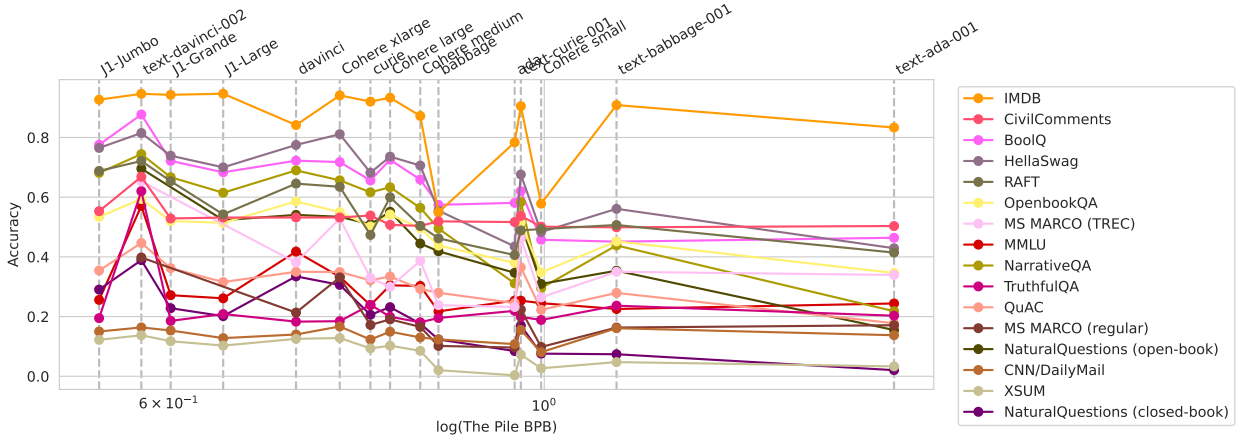

Figure 30: **The Pile loss vs. Accuracy.** The relationship between log bits-per-byte (BPB) on **The Pile** and the accuracy on each core scenario.